Khyber Pakhtunkhwa Public Service Commission (KPPSC)

Syllabus for Provincial Management Service (PMS) Competitive Examination

GEOGRAPHY — OPTIONAL SUBJECT (TOTAL MARKS: 200)

> Examination Structure: **Geography in KPPSC PMS is a full 200-mark optional subject divided into TWO distinct papers:**
> **- Paper-I: Physical and General Geography (100 Marks | 3 Hours)**
> **- Paper-II: Human, Economic and Regional Geography (100 Marks | 3 Hours)**
> **Examination Pattern for Both Papers: Attempt ANY FIVE (5) questions out of Eight (8). Each question carries 20 Marks. Time allowed: 3 Hours per paper.**

🌍 PAPER-I: PHYSICAL AND GENERAL GEOGRAPHY (100 MARKS)

Section 1: Elements of Weather and Climate (Climatology)

1. Insolation and Atmospheric Temperature:

- **Mechanism of solar insolation; Factors affecting solar radiation distribution (latitude, angle of incidence, duration of daylight, atmospheric absorption).**
- **Global radiation balance and terrestrial heat budget; Heat deficit and surplus zones; Latitudinal heat transfer.**
- **Horizontal and vertical distribution of temperature; Temperature inversion (types, causes, and environmental significance).**

2. Atmospheric Pressure and Wind Systems:

- **Planetary atmospheric pressure belts (Equatorial Low, Subtropical Highs, Subpolar Lows, Polar Highs).**
- **Mechanics of atmospheric circulation; Pressure gradient force, Coriolis force, frictional force, and geostrophic balance.**
- **Primary planetary wind systems (Trade Winds, Westerlies, Polar Easterlies); Secondary circulations (Monsoons, Land/Sea breezes, Anabatic/Katabatic winds).**

3. Humidity, Condensation, and Precipitation:

- **Atmospheric moisture: Absolute humidity, relative humidity, specific humidity, and dew point.**
- **Forms of condensation (clouds, fog, dew, frost); Cloud classification (high, middle, low, clouds of vertical development).**
- **Mechanisms of precipitation: Convectional, Orographic, and Cyclonic/Frontal precipitation; Global spatial distribution of precipitation.**

4. Air Masses, Fronts, and Atmospheric Disturbances:

- **Air masses: Definition, source regions, thermal and moisture classifications (cP, cT, mP, mT), and modification mechanisms.**
- **Frontogenesis and frontal types: Cold fronts, warm fronts, occluded fronts, and stationary fronts.**

- **Atmospheric disturbances: Life cycle and structure of mid-latitude wave cyclones (Polar Front Theory); Tropical cyclones (Hurricanes/Typhoons: origins, environmental conditions, eye wall mechanics, and track patterns); Anticyclones and associated weather.**

5. Climatic Classifications and World Climatic Regions:

- **Wladimir Köppen's climatic classification system: First-letter designations (A, B, C, D, E), second and third-letter criteria (precipitation seasonality and temperature characteristics).**
- **Detailed comparative analysis of major climatic types: Tropical Rainforest (Af), Tropical Savanna (Aw), Mediterranean (Cs), Marine West Coast (Cfb), Steppe (BS), and Desert (BW) climates.**

Section 2: Landforms and Geomorphology

1. Internal Structure of the Earth and Seismology:

- **Earth's interior layering: Crust (Sial and Sima), Mohorovičić discontinuity, Mantle (Asthenosphere and Mesosphere), Gutenberg discontinuity, Outer Core (liquid iron-nickel), and Inner Core (solid iron-nickel).**
- **Seismic wave behavior: Primary (P) waves, Secondary (S) waves, shadow zones, and seismic evidence for core composition and mantle rheology.**

2. Plate Tectonics and Tectonic Landforms:

- **Theory of continental drift (Wegener) and seafloor spreading (Hess); Plate Tectonics Paradigm.**
- **Three major types of plate boundaries and collisions: Divergent (rift valleys, mid-ocean ridges), Convergent (oceanic-continental, oceanic-oceanic, continental-continental orogenesis), and Transform faults.**
- **Earthquakes and Volcanicity: Mechanisms, epicenters, Wadati-Benioff zones; Volcanic landforms (intrusive: batholiths, laccoliths, sills, dykes; extrusive: shield volcanoes, composite cones, calderas); Tectonic and seismic zones of Pakistan (Chaman Fault, Main Boundary Thrust).**

3. Rocks and the Rock Cycle:

- **Definition and classification of rocks: Igneous (intrusive vs. extrusive, felsic vs. mafic mineral composition), Sedimentary (mechanically formed, chemically precipitated, organically formed), and Metamorphic (foliated vs. non-foliated, contact vs. regional metamorphism).**
- **The geological rock cycle and petrographic transformations.**

4. Geomorphic Processes: Endogenic and Exogenic Forces:

- **Weathering: Physical/mechanical weathering (frost wedging, thermal expansion/exfoliation, salt wedging) and Chemical weathering (carbonation, hydration, oxidation, hydrolysis); Biological weathering.**
- **Mass Wasting: Slope stability factors; Slow flowage (soil creep, solifluction), Rapid flowage (earthflow, mudflow, debris avalanche), Landslides (rockfall, rockslide, slumping).**

5. Fluvial, Glacial, Aeolian, Karst, and Coastal Systems:

- **Fluvial Geomorphology: Stages of a river (youth, maturity, old age); Erosional features (gorges, waterfalls, potholes, river capture); Depositional features (floodplains, meanders, oxbow lakes, braided streams, natural levees, deltas).**
- **Glacial Landforms: Valley/Alpine glaciers vs. Continental ice sheets; Erosional features (cirques, arêtes, horn peaks, U-shaped valleys, hanging valleys); Depositional features (moraines: terminal, lateral, medial, ground; drumlins, eskers, kames).**
- **Groundwater and Karst Topography: Prerequisites for Karst development; Surface features (sinkholes/dolines, uvalas, poljes, lapies); Underground features (caves, stalactites, stalagmites, columns).**

- **Aeolian Landforms: Wind erosion (deflation, abrasion, yardangs, zeugen); Wind deposition (dunes: barchans, seifs, transverse, longitudinal; loess deposits).**
- **Coastal Geomorphology: Wave action, longshore drift; Erosional coastal features (sea cliffs, wave-cut platforms, sea arches, stacks); Depositional coastal features (beaches, spits, bars, tombolos).**

6. Cycle of Erosion:

- **W.M. Davis's Geographical Cycle (Youth, Maturity, Old Age / Peneplain); Walther Penck's Morphological Analysis (Aufsteigende, Gleichförmige, Absteigende Entwicklung).**

Section 3: Oceans and Oceanography

1. Morphology of the Ocean Floor:

- **Major physiographic divisions: Continental shelf, continental slope, continental rise, deep ocean abyssal plains, oceanic trenches, mid-ocean ridges, and seamounts/guyots.**

2. Physical and Chemical Properties of Seawater:

- **Chemical composition: Salinity (average 35‰), major constituent salts (NaCl, MgCl2, etc.); Factors determining spatial and vertical salinity variations (evaporation, precipitation, river inflow, ice melting).**
- **Temperature distribution of ocean waters: Horizontal latitudinal variations, vertical thermal structure (thermocline layer).**

3. Ocean Water Movements: Waves, Currents, and Tides:

- **Oceanic Circulation: Major surface current gyres of the Atlantic Ocean (Gulf Stream, North Atlantic Drift, Canaries Current, Benguela), Pacific Ocean (Kuroshio, California, Humboldt/Peru Current), and Indian Ocean (Monsoon drift reversal).**
- **Tides: Equilibrium Theory (Newton) and Dynamic Theory (Laplace); Spring tides, neap tides, semi-diurnal, diurnal, and mixed tides; Tidal bores and coastal amplification.**

Section 4: Study of Maps and Cartography

1. Map Basics and Scales:

- **Definition and essentials of a map (title, orientation, scale, legend, grid/projection).**
- **Map scales: Statement of scale, Graphic/linear scale, Representative Fraction (RF: definition, mathematical calculation, merits, and demerits).**

2. Types of Maps:

- **Topographical maps (contour lines, relief representation, cultural features), Cadastral maps, Thematic maps.**
- **Synoptic Weather Maps: Construction principles, international meteorological symbols (isobars, fronts, precipitation, cloud cover), interpretation, and modern weather forecasting methods and constraints.**
- **Remote Sensing and Aerial Photography: Principles, stereoscopic viewing, photogrammetry, and GIS integration.**

3. Map Projections:

- **Definition, necessity of projections, and spherical distortion (shape, area, distance, direction).**
- **Classification: Cylindrical, Conical, Zenithal/Azimuthal, and Conventional projections.**
- **Properties, merits, and limitations: Mercator's Projection (rhumb line conformity vs. high-latitude areal exaggeration), Conical Projection with one/two standard parallels, Bonne's Projection, and**

Gnomonic Projection.

🌍 PAPER-II: HUMAN, ECONOMIC AND REGIONAL GEOGRAPHY (100 MARKS)

Section 1: Human Geography

1. Man and Habitat: Philosophical Paradigms:

- **Environmental Determinism: Friedrich Ratzel, Ellen Churchill Semple, Ellsworth Huntington.**
- **Possibilism: Paul Vidal de la Blache, Lucien Febvre.**
- **Neo-Determinism / Stop-and-Go Determinism: Griffith Taylor; Probabilism and Cultural Determinism.**

2. World Population Dynamics:

- **Spatial distribution and density of world population; Physical, cultural, and socio-economic controlling factors; Major ecumene and non-ecumene zones.**
- **Population Growth: Historical trajectory, J-shaped exponential curve, S-shaped logistic curve; Demographic Transition Model (5 stages); Malthusian Theory of Population vs. Neo-Malthusianism; Optimum Population Theory.**
- **Demographic Pressures in Pakistan: Population explosion, high dependency ratio, youth bulge, pressure on arable land, water resources, and urban infrastructure.**

3. Migration and Spatial Mobility:

- **Types of migration: Internal (rural-to-urban, seasonal) and International migration.**
- **Ravenstein's Laws of Migration; Lee's Push-Pull Model; Socio-economic and demographic consequences for sending and receiving regions (remittances, brain drain, urban slums).**

4. Human Settlements and Urbanization:

- **Rural Settlements: Site and situation factors, nucleated vs. dispersed settlements, pioneer settlement patterns.**
- **Urban Settlements: Definition, hierarchy, urban functions; Christaller's Central Place Theory (k=3 marketing, k=4 transport, k=7 administrative principles; threshold and range of a good).**
- **Urbanization: Global urbanization trends; Problems of rapid urbanization in developing countries (slums, katchi abadis, environmental degradation, water and sanitation deficits); Bridging the "green agenda" (ecological protection, open spaces) and "brown agenda" (safe water, sanitation, waste management) in sustainable urban planning.**

5. Society, Culture, and Geography of Religions:

- **Spatial distribution of major human races and language families.**
- **Geography of World Religions: Spatial distribution, historical diffusion, and geopolitical imprint of Islam and Christianity.**

Section 2: Economic Geography

1. Classification of Economic Activities:

- **Primary activities (agriculture, forestry, fishing, mining); Secondary activities (manufacturing, processing, construction); Tertiary activities (services, transport, trade); Quaternary**

(knowledge/information) and Quinary (high-level executive decision-making) sectors.

2. Agricultural Systems and Global Commodities:

- **Classification of agricultural systems: Subsistence vs. commercial, intensive vs. extensive agriculture, shifting cultivation, plantation agriculture, mechanized cereal farming, and mixed farming (livestock and crop integration).**
- **Major cereal and cash crops: Wheat, Rice, Cotton, and Sugarcane (geographical requirements, soil and climate, global production, spatial distribution, and international trade).**
- **Forest Resources: Tropical rainforests, tropical savanna woodlands, temperate coniferous forests; Economic benefits, deforestation, and environmental conservation.**
- **Marine and Freshwater Fisheries: Major commercial fishing grounds of the world (North-West Pacific/Japan, North-East Atlantic/Dogger Bank, North-West Atlantic/Grand Banks, South-East Pacific/Peru); Factors favoring commercial fishing.**

3. Power Resources and Energy Geopolitics:

- **Fossil fuels: Coal (types: anthracite, bituminous, lignite; major global coalfields), Petroleum / Crude Oil (formation, major oil-producing regions: Middle East, North America, Russia, offshore fields; OPEC and global trade), Natural Gas.**
- **Alternative and Renewable Energy: Hydroelectric power, nuclear, solar, wind, and geothermal energy; Energy security and geopolitics of energy pipelines.**

4. Industrial Location and Major Manufacturing Regions:

- **Theories of Industrial Location: Alfred Weber's Least-Cost Theory (Material Index, locational triangle, labor cost, agglomeration economies); August Lösch's Maximum Profit Location.**
- **Major World Industries: Iron and Steel Industry (locational shift from raw material to market/coastal sites; US Rust Belt, European Ruhr, Chinese industrial nodes); Textile Industry (natural vs. synthetic fibers; historical shifts); Chemical and Petrochemical Industries.**

5. Geography of Development and Global Economic Relations:

- **Indicators of Development: Economic (GDP, GNP, per capita income), socio-demographic (Human Development Index - HDI, Physical Quality of Life Index - PQLI, Gini Coefficient).**
- **Core-Periphery Model and Dependency Theory; Politico-economic relations between More Developed Countries (MDCs) and Less Developed Countries (LDCs); Regional economic blocs (EU, NAFTA/USMCA, ASEAN).**

Section 3: Regional Geography (South Asia with Special Focus on Pakistan)

1. Physiography and Drainage Base of Pakistan:

- **Physical divisions: Northern and Western Mountain Ranges (Karakoram, Himalayas, Hindu Kush, Safed Koh, Sulaiman, Kirthar); Potohar Plateau and Salt Range; Balochistan Plateau; Indus River Plain (Upper Indus Plain, Lower Indus Plain, Deltaic region).**
- **Drainage System: Indus River System and its eastern (Jhelum, Chenab, Ravi, Sutlej) and western tributaries (Kabul, Swat, Kurram, Gomal).**

2. Climatic Zones and Water Resources:

- **Climatic divisions of Pakistan (Highland, Lowland, Coastal, Arid); Monsoon circulation and Western Depressions.**
- **Water Resources and Irrigation Network: Canal irrigation system (inundation, perennial canals), dams and barrages; Indus Basin Treaty and hydrological challenges; Waterlogging, salinity, and climate change impacts.**

3. Agricultural Sector of Pakistan:

- **Structural characteristics of Pakistan's agro-economy; Kharif and Rabi cropping seasons; Major food crops (wheat, rice) and cash crops (cotton, sugarcane).**
- **Challenges facing agriculture: Land fragmentation, water scarcity, low per-acre yield, archaic practices, post-harvest losses, and climate volatility.**

4. Industrial Base and Mineral Wealth:

- **Major industries of Pakistan: Textiles, sugar, cement, fertilizer, steel, and chemical industries (locational factors, raw materials, domestic and export role).**
- **Mineral wealth: Gas fields (Sui), coal deposits (Thar coalfield), copper-gold reserves (Reko Diq, Saindak), rock salt, and limestone.**

5. Foreign Trade, Transport, and Geoeconomics:

- **Foreign trade patterns: Structural composition of exports (textiles, rice, leather, surgical instruments) and imports (petroleum, machinery, edible oil, chemicals); Persistent trade deficit causes and export expansion strategies.**
- **Transport infrastructure: National Highway Authority (NHA) network, Pakistan Railways, major seaports (Karachi Port, Port Qasim, Gwadar deep-sea port).**
- **Geoeconomic Paradigm & CPEC: Shift from geopolitics to geoeconomics; China-Pakistan Economic Corridor (CPEC) as the flagship corridor of the Belt and Road Initiative (BRI); Gwadar port connectivity, Special Economic Zones (SEZs), and regional transit potential for Central Asia.**
